(Please write your Enrollment Number) Enrollment No._____

MID-TERMEXAMINATION
(CBCS)
Course Name:B. Tech. (CSE- AI), Semester:3rd
(October, 2022)

Subject Code:BAI - 201 Subject: Artificial Intelligence
Time : 1 ½ Hours Maximum Marks : 30
Note: Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5*4)**

(a) Describe the salient features of an Agent.List the characteristic of intelligent Agent?

(b) Mention how Search strategies are Evaluated ?

(c) Determine whether the following two statements all logically equivalent

$\neg (P \rightarrow Q)$ and $(P \wedge \neg Q)$

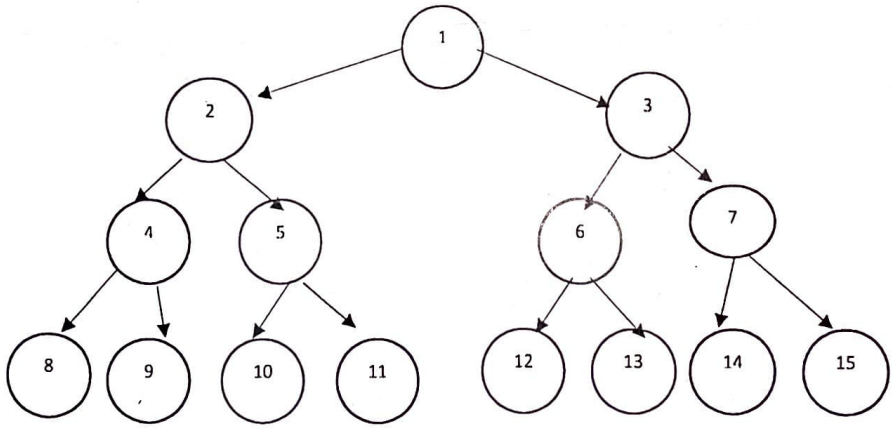
(c) What are the limitations in using propositional logic to represent the knowledge base?

Q2 State space for states 1 to 15 is given, suppose the goal state is 11. List the order in which nodes will be visited for: **(10)**

(a) Breadth-First search

(b) Depth limited Search with limit 3

(c) Iterative Deepening search (iterationwise)



Q3 Explain the A* search and give the proof of optimality of A* **(10)**

Q4 Translate these sentences into WFF and Clause form using Predicate Logic and prove that **John is Happy**using Resolution Algorithm. **(10)**

(a) Anyone passing his history exams and winning the Lottery is happy

(b) Anyone who studies or is Lucky can pass all his exams

(c) John did not study but he is Lucky

(d) Anyone who is lucky wins the Lottery

(e) There exists a person who is wealthy